

Практикум D7:archive

Seventeen questions, 91 marks. Every question the archive asks about a set of data, from May 2021 to November 2025 — summary numbers, box and whisker diagrams, correlation and regression — in the order of the eight techniques rather than the order of the papers.

О практикуме

statistics.regression целиком (33 блока, 69 баллов) и 22 балла описательной статистики из трёх тем вероятностей: 44 блока и 91 балл. Дубля нет — в теме нет ни одного блока ноября 2023. 79 баллов на Paper 2, 12 на Paper 1, Paper 3 нет вовсе.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d7-regression

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

D7 archive — describing data, all of it

Seventeen questions, 91 marks. Every question the archive asks about a set of data, from May 2021 to November 2025 — summary numbers, box and whisker diagrams, correlation and regression — in the order of the eight techniques rather than the order of the papers.

No theory. No worked examples. The theory is in the practicum, `practicum-d7-regression.ipynb`.

§	technique	parts	marks
1	The summary number	4	20
2	The box and its fences	3	13
3	The correlation coefficient	5	9
4	The line from the data	5	11
5	The prediction	7	15
6	Which line, and where not	2	10
7	The mean point	4	9
8	What the data do not say	4	4

The checks are the same ones the practicum uses and they store nothing. Each is handed the data exactly as the question gives them: `verify_fit` searches for the line with the least sum of squared misses, `verify_strength` takes r from the same search, `verify_missing` puts your values into the data and recounts, `verify_bound` solves the no-outlier inequality itself. Three significant figures, unless the question asks otherwise.

Parts of one question are split by technique. November 2021 Q1 appears in §§ 3, 4, 5 and 8; May 2021 TZ1 Q1 in §§ 4, 5 and 7. Each part repeats what it needs.

Four answers are words that nothing can compute — a reason, a claim, the name of a sampling method. Those are chosen from a list and checked by a hash.

Solutions are at the very bottom, deliberately far away.

```
import sys
sys.path.append('.')          # from
practicum/statistics to practicum/kit/
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *            # checks +
Sample, Box, Pairs, fit, strength

language('en')               # this
notebook is in English, and so are the checks

print('ready; sympy', sp.__version__)
```

§ 1. The summary number

Four questions, 20 marks. $\text{sum} = n \cdot \text{mean}$; the median is a place; σ divides by n .

1.1 — November 2025 TZ3 Paper 2 Q2, 6 marks

A teacher sets her class of 30 pupils a quiz. Aiden and Brett were absent. The box and whisker diagram of the 28 pupils who took the quiz has median 10.5, lowest score 6 and highest score 17. On their return Aiden scores less than 6 and Brett scores more than 17.

(a) Explain, briefly, why the median score for all 30 pupils would still be 10.5. Choose:

- 'a' — Aiden's score is below 10.5 and Brett's is above, so the middle value stays in the same place;
- 'b' — the mean only rises from 10.5 to 10.6;

- 'c' — the range of the 30 scores is 14;
- 'd' — adding two scores never changes a median.

The mean score of the 28 pupils was 10.5. The mean score for all 30 pupils is now 10.6. The range of scores for all 30 pupils is 14.

(b) Determine Aiden's score and Brett's score.

```
a, b = symbols('a b')

q1_la = ...      # 'a', 'b', 'c' or 'd'
q1_lb = ...      # (Aiden, Brett)

scores = Sample([Bunch(28, average=10.5,
lowest=6, highest=17), a, b])

check_word('1.1(a)', q1_la, 'ca978112ca1b')
verify_missing('1.1(b)', q1_lb, scores,
holds=[a < 6, b > 17], mean=10.6, range=14)
```

1.2 — May 2022 TZ1 Paper 2 Q1, 4 marks

exercising time (hours)	2	3	4	5	6
number of students	5	1	4	3	x

The median is 4.5 hours.

(a) Find the value of x .

(b) Find the standard deviation.

```
q1_2a = ...      # x
q1_2b = ...      # the standard deviation

hours = Sample({2: 5, 3: 1, 4: 4, 5: 3, 6:
x})

verify_missing('1.2(a)', q1_2a, hours,
median=4.5)
verify_summary('1.2(b)', q1_2b, hours.put({x:
7}), 'deviation')
```

1.3 — May 2024 TZ2 Paper 2 Q2(b), 4 marks

Consider the bivariate data set, where $p, q \in \mathbb{Z}^+$.

x	5	6	6	8	10
y	9	13	p	q	21

It is known that $\bar{y} = 16$. Given that $q - p = 3$, find the value of p and the value of q .

```
p, q = symbols('p q', positive=True)

q1_3 = ...      # (p, q)

verify_missing('1.3', q1_3, Sample([9, 13, p,
q, 21], 'y'), holds=[Eq(q - p, 3)], mean=16)
```

1.4 — November 2025 TZ1 Paper 2

Q3(a),(b), 6 marks

Data was collected on two variables, x and y , from eight members of a population.

x	3.8	6.3	2.5	2.4	8.9	10.7	6.7	10.3
y	11.8	9.7	6.1	4.1	p	23.3	13.7	24.5

(a) Find $\bar{x}$.

The regression line of y on x for this data is $y = 0.014375 + 2.1625x$.

(b) Find (i) $\bar{y}$; (ii) the value of p .

```
p = symbols('p')

q1_4a = ...      # x-bar
q1_4b_i = ...    # y-bar
q1_4b_ii = ...   # p

xs = Sample([3.8, 6.3, 2.5, 2.4, 8.9, 10.7,
6.7, 10.3])
ys = Sample([11.8, 9.7, 6.1, 4.1, p, 23.3,
13.7, 24.5], 'y')
line = Eq(y, 0.014375 + 2.1625*x)

verify_summary('1.4(a)', q1_4a, xs, 'mean')
verify_estimate('1.4(b)(i)', q1_4b_i, line,
mean(xs))
verify_missing('1.4(b)(ii)', q1_4b_ii, ys,
mean=0.014375 + 2.1625*mean(xs))
```

§ 2. The box and its fences

Three questions, 13 marks. $Q_3 + 1.5 \cdot \text{IQR}$ and $Q_1 - 1.5 \cdot \text{IQR}$; *no outliers* is an inequality.

2.1 — May 2021 TZ1 Paper 1 Q3, 5 marks

A box and whisker diagram of the weights of lizard eggs, in grams, has lowest value 10, lower quartile L , median 40, upper quartile U and highest value 75. The interquartile range is 20 grams and there are no outliers.

(a) Find the minimum possible value of U .

(b) Hence, find the minimum possible value of L .

```

L, U = symbols('L U')

q2_1a = ...      # the minimum U
q2_1b = ...      # the minimum L

eggs = Box(10, L, 40, U, 75)
rules = [Eq(iqr(eggs), 20), clean(eggs)]

verify_bound('2.1(a)', q2_1a, eggs, U,
'least', holds=rules)
verify_bound('2.1(b)', q2_1b, eggs, L,
'least', holds=rules)

```

2.2 — May 2022 TZ1 Paper 1 Q2(a), 3 marks

The ages of the eldest children of the adults at a swimming pool are summarised in a box and whisker diagram with lowest value 2, $Q_1 = 6$, median 7, $Q_3 = 10$ and highest value 18.

Find the largest value of c that would not be considered an outlier.

```

q2_2 = ...      # the largest c that is not
an outlier

verify_fence('2.2', q2_2, Box(2, 6, 7, 10,
18), 'upper')

```

2.3 — May 2022 TZ2 Paper 2 Q4(a)-(c), 5 marks

Nine adults had their reaction times measured after sleeping well. The box and whisker diagram reads: lowest 0.24, $Q_1 = 0.27$, median 0.28, $Q_3 = 0.35$, highest 0.46 (seconds).

(a) State the median reaction time after sleeping well.

(b) Verify that the measurement of 0.46 seconds is not an outlier.

(c) State why it appears that the mean reaction time is greater than the median reaction time.

(b): the upper fence, and 'yes' or 'no'.

(c): 'positive', 'negative' or 'none'.

```
q2_3a = ...           # the median
q2_3b_fence = ...     # the upper fence
q2_3b = ...           # is 0.46 an outlier?
q2_3c = ...           # the skew

slept = Box(0.24, 0.27, 0.28, 0.35, 0.46)

verify_summary('2.3(a)', q2_3a, slept,
               'median')
verify_fence('2.3(b) fence', q2_3b_fence,
             slept, 'upper')
verify_outlier('2.3(b)', q2_3b, slept, 0.46)
verify_skew('2.3(c)', q2_3c, slept)
```

§ 3. The correlation coefficient

Five questions, 9 marks. Three significant figures, and r , not r^2 .

3.1 — May 2021 TZ2 Paper 2 Q1(a)(ii), 3 marks with (a)(i)

number of customers (x)	3	9	11	10	5
waiting time (y minutes)	6	10	12	11	6

Write down the value of Pearson's product-moment correlation coefficient, r . (Part (a)(i), the line, is 5.2 below.)

```
q3_1 = ...      # r

verify_strength('3.1', q3_1, Pairs([3, 9, 11,
10, 5], [6, 10, 12, 11, 6]))
```

3.2 — November 2021 Paper 2 Q1(a), 2 marks

weekly practice (h hours)	28	13	45	33	17	29	39	36
diploma score (D)	115	82	120	116	79	101	110	121

Find Pearson's product-moment correlation coefficient, r , for these data.

```
q3_2 = ... # r

piano = Pairs([28, 13, 45, 33, 17, 29, 39,
36],
              [115, 82, 120, 116, 79, 101,
110, 121], names=('h', 'D'))

verify_strength('3.2', q3_2, piano)
```

3.3 — November 2022 Paper 2 Q1(b), 1 mark

Mathematics (x)	64	68	72	75	80	82	85	86
Science (y)	67	72	77	76	84	83	89	91

Write down the value of the Pearson's product-moment correlation coefficient, r .

```
q3_3 = ... # r

maths = Pairs([64, 68, 72, 75, 80, 82, 85,
86], [67, 72, 77, 76, 84, 83, 89, 91])

verify_strength('3.3', q3_3, maths)
```

3.4 — May 2024 TZ1 Paper 2 Q5(a), 2 marks

A, x	52	71	100	93	81	80	88	100	70	61
B, y	58	80	92	98	90	82	100	100	65	74

Find the value of Pearson's product-moment correlation coefficient, r .

```
q3_4 = ...          # r

tests = Pairs([52, 71, 100, 93, 81, 80, 88,
               100, 70, 61],
              [58, 80, 92, 98, 90, 82, 100,
               100, 65, 74])

verify_strength('3.4', q3_4, tests)
```

3.5 — November 2025 TZ1 Paper 2 Q3(c), 1 mark

For the data of 1.4, with the value of p found there, find Pearson's product-moment correlation coefficient, r .

```
q3_5 = ...          # r

data = Pairs([3.8, 6.3, 2.5, 2.4, 8.9, 10.7,
              6.7, 10.3],
             [11.8, 9.7, 6.1, 4.1, 18.5,
              23.3, 13.7, 24.5])

verify_strength('3.5', q3_5, data)
```

§ 4. The line from the data

Five questions, 11 marks. The pair (a, b) of $y = ax + b$, three significant figures.

4.1 — May 2021 TZ1 Paper 2 Q1(a), (d), 4 marks

x	3.3	6.9	11.9	13.4	17.8	19.6	21.8	25.3
y	6.3	8.1	8.4	11.6	10.3	12.9	13.1	17.3

The relationship between x and y can be modelled by the regression line of y on x with equation $y = ax + b$.

(a) Write down the value of a and the value of b .

(d) Draw the line of best fit on the scatter diagram.

For (d) enter the point, other than the y -intercept, that the line has to go through to earn the mark.

```

q4_1a = ...      # (a, b)
q4_1d = ...      # the point the line must
pass through

experiment = Pairs([3.3, 6.9, 11.9, 13.4,
17.8, 19.6, 21.8, 25.3],
                  [6.3, 8.1, 8.4, 11.6,
10.3, 12.9, 13.1, 17.3])

verify_fit('4.1(a)', q4_1a, experiment)
verify_centre('4.1(d)', q4_1d, experiment)

```

4.2 — November 2021 Paper 2 Q1(b), 1 mark

For the practice data of 3.2, the relationship can be modelled by the regression equation $D = ah + b$. Write down the value of a and the value of b .

```

q4_2 = ...      # (a, b)

verify_fit('4.2', q4_2, piano)

```

4.3 — November 2022 Paper 2 Q1(a), 2 marks

For the Mathematics and Science scores of 3.3, the regression line of y on x can be written in the form $y = ax + b$. Find the value of a and the value of b .

```
q4_3 = ...          # (a, b)

verify_fit('4.3', q4_3, maths)
```

4.4 — May 2023 TZ2 Paper 2 Q1(a), 2 marks

A botanist measures the average height, h cm, of plants after d days.

days (d)	2	5	13	24	33	37	42
height (h)	10	16	30	59	76	79	82

The regression line of h on d can be written $h = ad + b$. Find the value of a and the value of b .

```
q4_4 = ...          # (a, b)

plants = Pairs([2, 5, 13, 24, 33, 37, 42],
               [10, 16, 30, 59, 76, 79, 82], names=('d',
               'h'))

verify_fit('4.4', q4_4, plants)
```

4.5 — May 2025 TZ2 Paper 2 Q1(a), 2 marks

play time (x hours)	11	13	14	17	22	24
sleep time (y hours)	62	65	68	75	84	87

The regression line of y on x can be written $y = ax + b$. Find the value of a and the value of b .

```
q4_5 = ...          # (a, b)

sleep = Pairs([11, 13, 14, 17, 22, 24], [62,
65, 68, 75, 84, 87])

verify_fit('4.5', q4_5, sleep)
```

§ 5. The prediction

Seven questions, 15 marks. Substitute into the line; a is the change per unit.

5.1 — May 2021 TZ1 Paper 2 Q1(b), 2 marks

Use the model of 4.1 to predict the value of y when $x = 18$.

```
q5_1 = ...          # y at x = 18  
  
verify_estimate('5.1', q5_1, experiment, 18)
```

5.2 — May 2021 TZ2 Paper 2 Q1(a)(i), (b),(c), 3 marks

For the café data of 3.1, with the regression line $y = ax + b$ of waiting time on number of customers:

- (b)** Interpret, in context, the value of a .
- (c)** Seven customers are already waiting. Estimate Sarah's waiting time.

(b): the number of minutes the waiting time changes for one more customer.

```

q5_2b = ...          # minutes per one more
customer
q5_2c = ...          # the waiting time with 7
customers ahead

cafe = Pairs([3, 9, 11, 10, 5], [6, 10, 12,
11, 6])

verify_change('5.2(b)', q5_2b, cafe, 1)
verify_estimate('5.2(c)', q5_2c, cafe, 7)

```

5.3 — November 2021 Paper 2 Q1(c), 2 marks

One of the students of 3.2 wished she had practised more. Based on the given data, determine how her score could have been expected to alter had she practised an extra five hours per week.

```

q5_3 = ...          # the change in the score

verify_change('5.3', q5_3, piano, 5)

```

5.4 — November 2022 Paper 2 Q1(c), 2 marks

Use the regression line of 4.3 to predict the Science test score for a student who has a score of 78 on the Mathematics test, to the nearest integer.

```
q5_4 = ...          # the Science score

verify_estimate('5.4', q5_4, maths, 78,
whole=True)
```

5.5 — May 2023 TZ2 Paper 2 Q1(b), 2 marks

Use the regression line of 4.4 to estimate the average height of the plants when the experiment has been running for 20 days.

```
q5_5 = ...          # h at d = 20

verify_estimate('5.5', q5_5, plants, 20)
```

5.6 — May 2025 TZ1 Paper 2 Q4(a), 2 marks

For a large group of adults the regression line of foot length on arm span is $F = 0.335A - 32.6$, and of arm span on foot length $A = 2.89F + 99.3$. By using an appropriate regression line, find an estimate of the arm span for an adult with a foot length of 19.8 cm.

```
q5_6 = ...          # the arm span

A, F = symbols('A F')
f_on_a = Eq(F, 0.335*A - 32.6)
a_on_f = Eq(A, 2.89*F + 99.3)

verify_estimate('5.6', q5_6, [f_on_a,
a_on_f], 19.8, of=A)
```

5.7 — May 2025 TZ2 Paper 2 Q1(b), 2 marks

Use the regression line of 4.5 to estimate the sleep time of a child whose weekly play time is 20 hours.

```
q5_7 = ...          # the sleep time

verify_estimate('5.7', q5_7, sleep, 20)
```

§ 6. Which line, and where not

Two questions, 10 marks. Predict x with the line of x on y ; outside the data, predict nothing.

6.1 — May 2023 TZ1 Paper 2 Q3(b), (c), 5 marks

The number of children in a park is modelled by $y = -0.6T^2 + 23T + 110$, where T °C is the highest temperature. An ice cream vendor records, on five days:

children (y)	81	175	202	346	360
ice creams sold (x)	15	27	23	35	46

(b) Find an appropriate regression equation that will allow the vendor to predict the number of ice creams sold on a day when there are y children.

(c) Hence predict the number of ice creams sold on a day when the highest temperature is 25 °C.

```

q6_lb = ...      # the regression equation
q6_lc = ...      # ice creams, a whole number

park = Pairs([81, 175, 202, 346, 360], [15,
27, 23, 35, 46], names=('y', 'x'))

verify_fit('6.1(b)', q6_lb, park, of='x')
verify_estimate('6.1(c)', q6_lc, park,
-0.6*25**2 + 23*25 + 110, of='x', whole=True)

```

6.2 — May 2024 TZ1 Paper 2 Q5(b), (c), 5 marks

For the tests of 3.4 the regression line of y on x is $y = 0.822x + 18.4$. Paulo scored 10 on Test A and was absent for Test B; the teacher estimated his Test B score from this line. Giovanni scored 90 on Test B and was absent for Test A; the teacher solved $90 = 0.822x + 18.4$ for his Test A score.

(b) Give a reason why this method is not appropriate for Paulo.

(c) (i) Give a reason why this method is not appropriate for Giovanni. (ii) Use an appropriate method to estimate Giovanni's Test A score to the nearest integer.

Reasons: 'extrapolation' or 'wrong line'.

```
q6_2b = ...      # the reason for Paulo
q6_2c_i = ...    # the reason for Giovanni
q6_2c_ii = ...   # Giovanni's Test A score

verify_reason('6.2(b)', q6_2b, tests, 10,
given='x', predict='y')
verify_reason('6.2(c)(i)', q6_2c_i, tests,
90, given='y', predict='x')
verify_estimate('6.2(c)(ii)', q6_2c_ii,
tests, 90, of='x', whole=True)
```

§ 7. The mean point

Four questions, 9 marks. Every regression line passes through $(\bar{x}, \bar{y})$; two lines meet there.

7.1 — May 2021 TZ1 Paper 2 Q1(c), 1 mark

For the data of 4.1, write down the value of $\bar{x}$ and the value of $\bar{y}$.

```
q7_1 = ...          # (x-bar, y-bar)

verify_centre('7.1', q7_1, experiment)
```

7.2 — May 2024 TZ2 Paper 2 Q2(a), 1 mark

The regression line of y on x for the data of 1.3 is $y = 2.1875x + 0.6875$, and it passes through the mean point. Given that $\bar{x} = 7$, verify that $\bar{y} = 16$: enter the value the line gives.

```
q7_2 = ...          # y-bar

verify_estimate('7.2', q7_2, Eq(y, 2.1875*x + 0.6875), 7)
```

7.3 — May 2025 TZ1 Paper 2 Q4(b), 3 marks

For the adults of 5.6, find the mean arm span and the mean foot length.

The pair (mean arm span, mean foot length).

```
q7_3 = ...          # (mean A, mean F)

verify_centre('7.3', q7_3, [f_on_a, a_on_f],
order=(A, F))
```

7.4 — May 2022 TZ1 Paper 1 Q2(b), 4 marks

At the swimming pool of 2.2 the age of an adult is a years and of their eldest child c years. The regression line of a on c is $a = \frac{7}{4}c + 20$; the regression line of c on a is $c = \frac{1}{2}a - 9$.

(i) One of the adults surveyed is 42 years old. Estimate the age of their eldest child.

(ii) Find the mean age of all the adults surveyed.

```
q7_4_i = ...      # the child's age
q7_4_ii = ...     # the mean age of the adults

a, c = symbols('a c')
a_on_c = Eq(a, Rational(7, 4)*c + 20)
c_on_a = Eq(c, a/2 - 9)

verify_estimate('7.4(i)', q7_4_i, [a_on_c,
c_on_a], 42, of=c)
verify_centre('7.4(ii)', q7_4_ii, [a_on_c,
c_on_a], of=a)
```

§ 8. What the data do not say

Four questions, 4 marks. Correlation is not cause; r ignores shifts.

8.1 — *November 2021 Paper 2 Q1(d), 1 mark*

Lucy asserts that the number of hours a student practises (3.2) has a direct effect on their final diploma result. Comment on the validity of her assertion:

- 'a' — valid: r is close to 1;
- 'b' — not valid: the data can only show a correlation, not a cause;
- 'c' — valid: the gradient of the line is positive;
- 'd' — not valid: eight students are too few for any regression.

```
q8_1 = ...          # 'a', 'b', 'c' or 'd'

check_word('8.1', q8_1, '3e23e8160039')
```

8.2 — November 2021 Paper 2 Q1(e), 1 mark

Lucy deducts a fixed number of hours per week from each student's recorded hours. State how, if at all, the value of r would be affected: 'no effect', 'increases' or 'decreases'.

```
q8_2 = ...          # the effect on r

verify_effect('8.2', q8_2, piano, lambda
hours: hours - 3, which='h')
```

8.3 — May 2022 TZ2 Paper 2 Q4(d), 1 mark

After **not** sleeping well, the nine adults of 2.3 have the box and whisker diagram 0.27, 0.29, 0.35, 0.39, 0.48. Comment on whether the two diagrams provide any evidence that might suggest that not sleeping well causes an increase in reaction time:

- 'a' — yes: the median after not sleeping well equals the upper quartile after sleeping well;

- 'b' — no: correlation does not imply causation;
- 'c' — yes: the mean reaction time is higher after not sleeping well;
- 'd' — no: the two diagrams overlap.

```
q8_3 = ...          # 'a', 'b', 'c' or 'd'  
  
check_word('8.3', q8_3, 'ca978112ca1b')
```

8.4 — May 2025 TZ3 Paper 2 Q3(c), 1 mark

A supermarket manager surveys the first 50 customers to arrive on a particular day. Which one best describes the sampling method: simple random, systematic, convenience, quota or stratified?

```
q8_4 = ...          # the sampling method  
  
check_word('8.4', q8_4, 'd7ae368c1499')
```



Solutions

§ 1

1.1 (a) $\boxed{a}$: the median is the middle value, and as Aiden's score is below 10.5 and Brett's is above, the middle stays in the same place.

1.1 (b) $30 \cdot 10.6 - 28 \cdot 10.5 = 318 - 294 = 24$ for the two together; the new range is Brett's minus Aiden's, 14. So Aiden $\boxed{5}$, Brett $\boxed{19}$.

1.2 (a) Ten values up to 4, so $\Sigma f = 20$: $x = \boxed{7}$. **(b)** $\sigma = \boxed{1.58}$ (1.58429...).

1.3 $\frac{9+13+p+q+21}{5} = 16$ gives $p + q = 37$; with $q - p = 3$: $p = \boxed{17}$, $q = \boxed{20}$.

1.4 (a) $\bar{x} = \frac{51.6}{8} = \boxed{6.45}$. **(b)(i)** $\bar{y} = 0.014375 + 2.1625 \cdot 6.45 = 13.9625 \approx \boxed{14.0}$. **(ii)** $8 \cdot 13.9625 = 111.7 = 93.2 + p$, so $p = \boxed{18.5}$.

§ 2

2.1 (a) $75 \leq U + 30$ gives $U \geq 45$ (and the lower fence, $10 \geq U - 50$, allows up to 60): $U = \boxed{45}$. **(b)** $L = U - 20 = \boxed{25}$.

2.2 $10 + 1.5 \cdot (10 - 6) = \boxed{16}$.

2.3 (a) $\boxed{0.28}$ s. **(b)** $0.35 + 1.5 \cdot 0.08 = \boxed{0.47}$ and $0.46 < 0.47$: $\boxed{\text{not an outlier}}$. **(c)** The median is closer to Q_1 : $\boxed{\text{positively skewed}}$, so the mean is pulled above the median.

§ 3

3.1 $r = \boxed{0.978}$. **3.2** $r = \boxed{0.884}$. **3.3** $r = \boxed{0.981}$. **3.4** $r = \boxed{0.901}$. **3.5** $r = \boxed{0.954}$ ($0.953958\dots$).

§ 4

4.1 (a) $a = \boxed{0.433}$, $b = \boxed{4.50}$. **(d)** Through $\boxed{(15, 11)}$, the mean point, and meeting the y -axis near 4.50.

4.2 $a = \boxed{1.37}$, $b = \boxed{64.5}$. **4.3** $a = \boxed{1.01}$, $b = \boxed{2.45}$. **4.4** $a = \boxed{1.93}$, $b = \boxed{7.22}$

$(1.93258\dots, 7.21662\dots)$. **4.5** $a = \boxed{1.98}$, $b = \boxed{40.2}$ $(1.97651\dots, 40.2286\dots)$.

§ 5

5.1 $\boxed{12.3}$. **5.2 (b)** An increase of $\boxed{0.805}$ minutes of waiting time per additional customer ahead. **(c)** $\boxed{8.52}$ minutes.

5.3 $5 \cdot 1.36609\dots = \boxed{6.83}$: about seven marks more.

5.4 $81.39\dots \approx \boxed{81}$. **5.5** $\boxed{45.9}$ cm $(45.8683\dots)$.

5.6 Use A on F : $2.89 \cdot 19.8 + 99.3 = 156.522 \approx \boxed{157}$ cm.

5.7 $\boxed{79.8}$ hours $(79.7589\dots)$.

§ 6

6.1 (b) Predicting x from y : the line of x on y , $\boxed{x = 0.0935y + 7.43}$. **(c)** $y(25) = 310$ and $0.0935114\dots \cdot 310 + 7.43053\dots = 36.4\dots \approx \boxed{36}$.

6.2 (b) 10 is outside the Test A data: extrapolation. **(c)(i)** Predicting x with the line of y on x : the wrong line. **(ii)** The line of x on y at $y = 90$ gives $85.62 \dots \approx$ 86.

§ 7

7.1 (15, 11). **7.2** $2.1875 \cdot 7 + 0.6875 =$ 16.

7.3 The lines meet at the mean point: $\bar{A} =$ 160 cm, $\bar{F} =$ 20.9 cm.

7.4 (i) $c = \frac{1}{2} \cdot 42 - 9 =$ 12. **(ii)** $a = \frac{7}{8}a + \frac{17}{4}$, so $\bar{a} =$ 34.

§ 8

8.1 b. **8.2** no effect. **8.3** a — R0 for *correlation does not imply causation*, and a comparison of means is not accepted. **8.4** convenience.